
Algorithm 1: One calibration round as a dual optimization. The human concept H_t and the external annotation policy Θ_t improve together: the machine proposes and pre-labels under seal, the person rules blind, and only then is the seal opened so the disagreement can diagnose the policy rather than replace the judgment.

Input: closed G_{t-1} , closed D_{t-1} , development pool, human capacity m_t

Output: closed D_t , closed G_t , stop verdict

```
1 ▷ the machine proposes
2  $C_t \leftarrow$  candidates drawn across the seven diagnostic regions
3  $P_t \leftarrow$  independent weak-executor prelabels under  $G_{t-1}$ , sealed
4  $B_t \leftarrow$  challenge items  $\cup$  stratified consensus audit, within  $m_t$ 

5 ▷ the human rules, blind to  $P_t$ 
6 foreach item  $i \in B_t$  do
7   |  $Y_t^*(i) \leftarrow$  human class, region, uncertainty, rationale
8 end
   //  $P_t$  opens only after  $Y_t^*$  is fixed, so it can diagnose the guideline
   without ever standing in for gold

9 ▷ both learners adapt
10  $H_t \leftarrow H_{t-1}$  sharpened by the comparisons just made
11  $\Theta_t \leftarrow$  revise  $\Theta_{t-1}$  to express  $H_t$ , using  $P_t$  as error evidence

12 ▷ checkpoint
13  $D_t \leftarrow D_{t-1} \cup Y_t^*$ 
14  $G_t \leftarrow$  close  $\Theta_t$  as a version
15 if  $STOPGATE(D_t, G_t)$  holds then
16   | return  $D_t, G_t$ , stop                                //  $G^* \leftarrow G_t$ ; the sealed test may open
17 else
18   | return  $D_t, G_t$ , continue                                //  $t \leftarrow t + 1$ 
19 end
```
